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Research Program/Project/Study Title: Development of Capacity Enhance Cold Water Dispenser	
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas City						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☒ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☒ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☒ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								

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(7) Sector/Commodity

Engineering and Construction Firm
Hardware & Supplies
Automotive shop

(8) Discipline/Field of Specialization

HVAC-R Technology
Drafting Technology
Architecture
Automotive Technology

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

A researchers develop the Development of Capacity Enhance Cold Water Dispenser for research study with the desire to improve, enhance the usefulness and practical effectivity, As well as y the consumer needs and for economic development.

In this problem, a study that somehow can be used as an Development of Capacity Enhance Cold Water Dispenser as well as performance machine aparatus. To make research, do some experiments, share ideas, knowledge, and thoughts in able to develop Development of Capacity Enhance Cold Water Dispenser : By this information stated, it can help and improve the needs of individuals . This study enhances the performance level of every consumer who wants to use this product.

A kind of device can be use to provide ease of the demand of consumers needs in terms of dispensing capacity of the device in terms of temperature. The essence of this research is to add feature of the device and mobility for workmanship, it will enhance the consumers capability to provide cold water using Development of Capacity Enhance Cold Water Dispenser . Perhaps it is difficult as first time to introduce this kind of equipment but to maintain the ability of innovative mind when it comes to technical specification, performance and applicability surely we can able to adapt and fabricate Development of Capacity Enhance Cold Water Dispenser .Objectives of the Study: Generally, This study will be conducted in order to develop a Development of Capacity Enhance Cold Water Dispenser . Specifically, it will seek to determine the following:

1. Determine the effeciency of Development Capacity Enhance Cold Water Dispenser interms of dispensing capacity rate.
2. Determine the level of acceptability of Development Capacity Enhance Cold Water Dispenser in terms.
 - a.design
 - b. fabrication
 - c. performance of the machine

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The design is actually the product of continuous effort of the researchers to come up with a device suitable for household and commercials.

In designing, the researcher will be developing a Development of Capacity Enhance Cold

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Water Dispenser with, electrical/electronics components and mechanical parts which is properly joined and mechanically fastenincludes their specific functions. The design is composed of three main parts which utilized the mechanical, electrical and electronic control systems. It also include body frame and caster wheel with lock whenever the device is used. The device can be operational and functional according to each parts and it will be done through domestic technology information

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Prior Arts

Compressor has similar function as identified in this study. It is divided into several types (depending on the size of the plant, the refrigerant they use, and other factors) including single-cylinder or multi-cylinder. With respect to the method of compressing, they are classified as reciprocating, notary or centrifugal. They may employ either direct drive or belt drive with respect the location of the prime mover. They are classified as either independent (belt drive) or semi-hermetic (direct drive, motor and compressor in separate housing).

This is how it works. The passage of the refrigerant to and from the compressor is controlled by means of a discharge and suction value located on a specially designed value plate, which form the lower part of the cylinder head. The flapping action of the value permits the flow of refrigerant out only through the discharged value port and flow in only through the suction valve port.

Thus, when the piston move away from the value plate (suction stroke), a pressure reduction takes place because the pressure in the cylinder is now below the suction line, a flow of refrigerant occurs that pushes, open the suction valve and permits a certain quality of refrigerant to enter the compressor.

As the motion of the piston reverse and moves forward, the valve plate (compressor stroke), increases the pressure, which forces the suction valve to close. Further compression occurs as the piston moves close to the valve plate and opens the discharge valve, which forces the refrigerant into the discharge line (thus causing what is known as the high-side pressure or positive working pressure of the refrigerant).

Reciprocating compressors on the other hand derive their name from the reciprocating (back–and–forth) action of the piston. The length of the piston movement within its cylinder is known as the stroke. The capacity of the compressor depends on such factors as the number of cylinders, strokes, and revolutions per minute of the crankshaft (a modern type of compressor with the electric motor enclosed in a hermetically sealed compressor machinery) a heavy duty two cylinder industrial compressor or vertical design and low-speed operation. The motion of the piston is transferred from the crankshaft which, in turn, receives its motion from a belt wheel. the prime mover for this type of compressor is usually an electric motor, although there is no reason why other types of prime movers cannot be used.

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The piston motion is attached by the crankshaft arm (or arms), which is fastened to one end of a connecting rod, with the other end of the connecting rod fasted to the piston by means of piston pin. In this manner the connecting rod and crank arm transferred the rotation of the crankshaft to the reciprocating or back and forth motion of the piston.

In addition to reciprocating compressors, hermetic (or sealed type) compressor are directly connected to an electric motor. The motor and compressor operate on the same shaft and are enclosed in a common casing. Condensing units of this type are used almost exclusively in domestic refrigerators and also in locker and home cold-storage plants, soda fountain, ice cream and food display cabinet, drinking fountain and the like. They are made to operate on either the reciprocating or the rotary principled and may be mounted with the shaft. In either the vertical or the horizontal position in a unit of this type, the revolutions per minute (rpm) obviously are the same for both compressor and motor, this factor has a very important bearing on the size and design of the unit since it determined the type of refrigerant, the type of controls to used, and so on.

In a unit of this type the one piece housing provides for quietness and a minimum of vibration. In addition, the seal and coupling (always a maintenance problem in open compressors) are eliminated. Still another dependable feature is the fact that it is entirely enclosed, no airborne dust or dirt can react on it. Suction gas at 50°K to 60°F cools the motor and shell. Together, the foregoing factors ensure long, trouble-free motor operation.

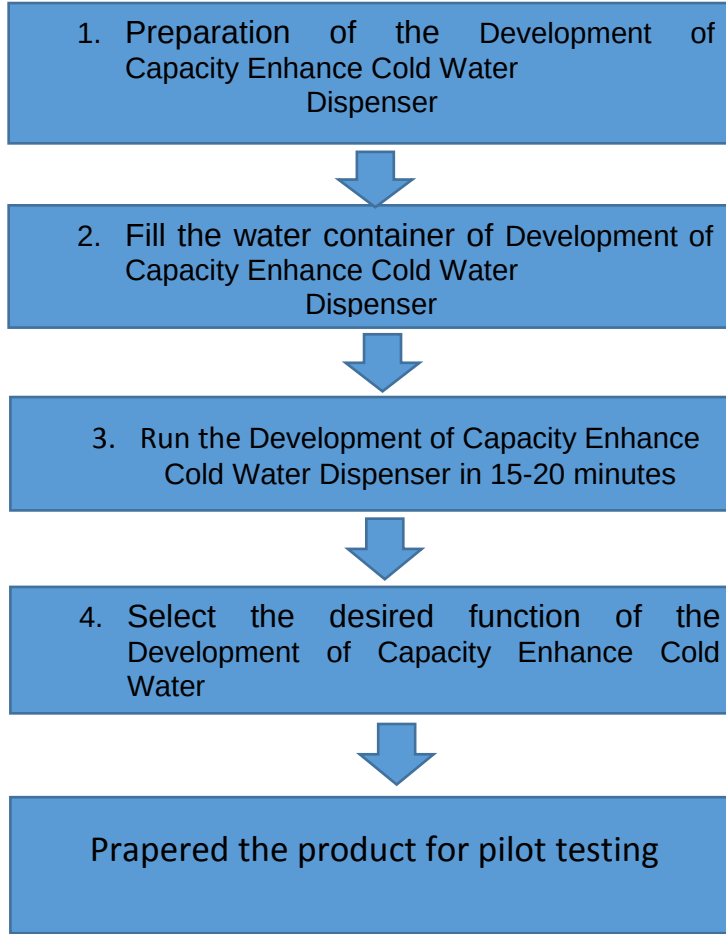
This particular unit is internally spring mounted. The motor, located above the compressor, operates in a vertical position, whereas the compressor in oil, simplifying the lubricating problem. The suction intake is placed so that the suction vapor must travel through the holes in the motor rotor to get the top of the shell and them to the intake tube.

Certain models of this type of compressor are provided with internal thermostats that are inserted in the motor windings. Therefore, measure motor temperature exactly, without allowing for the air gap between the motor and the top of the shell, where overloads are normally located. This is particularly important in heat-pump application where the ambient temperature may have considerable influence on the protection system. Because the thermostat is located all the most critical point, it gives instantaneous and accurate sensing of the motor temperature and, therefore, can remove the compressor from the line at a safe temperature level. It is always operative when the compressor is running.

Another important feature is an antilog device, consisting of two assemblies. One is the centrifuge, which is press fit on the crankshaft and therefore, rotates at the speed of the compressor. The refrigerant is drawn in through the holes on the top. Any liquid or oil expelled through the slots on the side by centrifugal force, and the gas (being lighter) is drawn through the slots in the hub the second assembly collects the gas and directs it to the cylinder heads. This system always operates when the compressor is running and functions under all conditions that may affect slugging. It is not dependent upon any external component, which may fail, and so is practically foolproof.

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(12) Scientific Basis/Theoretical Framework(Include concepts, theories and variables that indicate relevant relationships to be tested.



Flow Chart of Development of Capacity Enhance Cold Water Dispenser

(13) Methodology(Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Methodology:

This study relates to the actual performance of commercial water dispensers in producing cold water.

This innovation is to provide a Development of Capacity Enhance Cold Water Dispenser that will enhances the ability of the students to practice the actual servicing and repair of refrigeration and air conditioning. This equipment must encourage the instructors and professors of this field of specialization to use the skills acquired and earn money out of the product made. They can more effective RAC technician.

This device features positive in operation, convenient in use, easily installed in a working position, and easily dis-connected, economically constructed, relatively simple, and of general

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superiority and serviceability (Priestly,1962).

Design Plan and Preparation

The design is actually the product of the continuous effort of the researcher to come up with a device suitable for the Development of Capacity Enhance Cold Water Dispenser module.

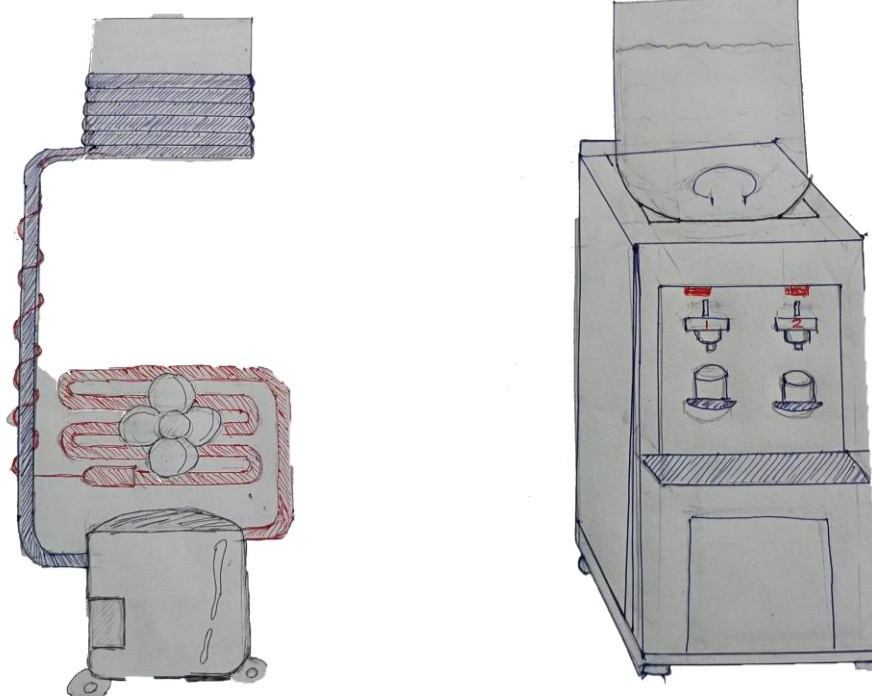
In designing, the researcher develops development of Capacity Enhance Cold Water Dispenser having its technical specification, operating performance, and applicability in the workplace. The device is composed of different parts which utilized the Development of a Capacity Enhance Cold Water Dispenser system with dual switching devices for the operating system.

Operation of Development of Capacity Enhance Cold WaterDispenser

The structural parts used for the operation were prepared. Then, fill the water container to utilize for storing potable water. Electronics switching devices are responsible for the auto-switching function. Before using, the Development of Capacity Enhance Cold Water Dispenser structural, select the desired temperature then plug in the unit and turn on the device in 15-20 minutes before use.

Fabrication of Development of Capacity Enhance Cold WaterDispenser

The device will be fabricated with the following procedure. Development of Capacity Enhance Cold Water Dispenser. The parts and components that are fastened will be brazed, and all the tubing system of mechanical parts of the refrigeration system with a charge of refrigerant. Attached are all the electronic parts and switching devices. electronic temperature controlled is set accordingly for standard setting temperature.



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Evaluation Procedure

Testing of the operating performance of the Development of Capacity Enhance Cold Water Dispenser

Assembling the device in accordance with the appropriate specification, different parts and functions will be able to form the extended length and height of the Development of Capacity Enhance Cold Water Dispenser. The main frame or structure is being utilized where the body frame and enclosed equipment are attached for different uses. The refrigeration system is responsible for the cooling effect and the electronic switching devices are responsible for the automatic switching effect and dispensing the water cold the devices are also circuit breaker protection devices to insure the safety of the users.

Evaluation of the output or finished product

The finished product will be evaluated by twenty (20) evaluators who came from academe composed of Industrial Technology and Engineering professors. The equipment will be evaluated as to its performance like usefulness, capacity, and safety feature. The equipment will be further evaluated based on the technical specification/feature and operating performance using a researcher-made evaluation sheet.

Instrumentation and Data Gathered

The evaluation of the product will be done using an evaluation sheet which will be formulated by the researcher. It will be validated by the panel of experts and the members. The evaluation sheet will be pre-tested among evaluators who have the same qualification as the evaluators. The evaluators will be composed of professionals (electronics, electrical, HVAC, and automotive professor) and students who will evaluate the Development of Capacity Enhance Cold Water Dispenser in terms of dispensing capacity of cold water, and quantity temperature in a short period of time.

To Determine the level of acceptability of Development Capacity Enhance Cold Water Dispenser in terms of technical specification, safety, and composition.

The evaluation sheet will be the main data-gathering instrument of this study. It is composed of measuring test parameters considered in the study. A five (5) point scale will be used to respond to categories and the acceptability of the mechanism, functionality, and operating procedure.

Scale of Means	Verbal Interpretation
4.21-5.00	Very Efficient
3.41-4.20	Efficient
2.61-3.40	Moderately Efficient
1.81-2.60	Less Efficient
1.0-1.80	Least Efficient

The data gathering will be done through the actual observation and manipulation in determining the quantitative operating performance and acceptability of the Development of Capacity Enhance Cold Water Dispenser. The accomplished evaluation sheets will be gathered and consolidated for scientific analysis and interpretation purposes.

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Parameters for Analysis

A developmental study of Development of Capacity Enhance Cold Water Dispenser. Electronics/Electrical being installed for operation with the built-in timer. While cooling temperature the temperature controlled is attached to secure the cooling effect of the machine.

To ascertain the performance and acceptability of the Development of Capacity Enhance Cold Water Dispenser, the following parameters were considered:

1. Mechanism
2. Functionality
3. Operating Procedure

This research study developed an instrument to test the specific objective of the quality result of the Development of Capacity Enhance Cold Water Dispenser.

This is also to advertise the advantages of using this equipment in extension services. Such the needs of consumers in such quality temperatures are produced to answer the needs of the community.

(15) Expected Output(Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Development of Capacity Enhance Cold Water Dispenser will be fabricated with metal framing and brackets to ensure the quality and durability of the products, the fabricated tubing system is sought to the standard quality of materials, and the mechanical parts of the refrigeration system are used. The additional feature of the device is having a dual switching device to operate domestically and commercially, the product can produce the desired temperature needed by the consumers.

(15) Technology Transfer Activity Design(Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The product will be used in domestic and commercial establishments in order to produce more capacity cold water to the consumer needs.

This enables a person to use it with proper orientation so that it will enhance the ability to simplify technical specifications/features of the device were could able to practice ease of work. It will also be utilized in order to produce more capacity of cold water effectively and to maintain the required temperature. The device will be used in domestic and commercial establishments in order to use the device effectively as development of the device in such a way it will provide quality water dispensing. This device will be introduced to the extension services for alternative learning and can be a way of making a device as a source of income for their own personal needs.

(16) Targets Beneficiaries(Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The Development of Capacity Enhance Cold Water Dispenser will eventually benefit

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the following:

HVAC Students taking up Heating Ventallating Air conditioning Refrigetaion Technology course who are particular in doing actual learning activity.

Professors with their field of specialization in HVAC.

Manufacturer those who fabricate tools and equipment will benefit from this product. The product can also be introduced to the consumer for producing Capacity Enhance Cold Water Dispenser to the needs of consumers satisfactions.

Institutions and Universities those who offer HVAC can recommend for the utilization of the device, for techno-transfer, promotion and commercialization.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		
- Meal allowance during data gathering		
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		
- Printing		
C. CAPITAL OUTLAY		
Total		

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

“Webpage of x-rite, manufacturer of spectrophotometers,” 2017,[Accessed 21-May-2017]. [Online]. Available: <http://www.xrite.com/products?industry=RetailPaint&sort=4&pagesize=6>, X-Rite,

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› PDF System Units Cleaning Reference Guide. Revised: October 21, 2020

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Cost and materials

<i>materials</i>	<i>unit</i>	<i>quantity</i>	<i>prices</i>
<i>Freon 134a</i>	<i>1 kilo</i>	<i>1</i>	<i>500</i>
<i>Silver rod</i>	<i>5 pcs.</i>	<i>5</i>	<i>100</i>
<i>Copper tube</i>	<i>1 roll</i>	<i>1</i>	<i>1600</i>
<i>compressor</i>	<i>1 pc</i>	<i>1</i>	<i>500</i>
<i>Coin slots ammbles</i>	<i>2 sets</i>	<i>2</i>	<i>2500</i>
<i>Ball caster</i>	<i>4 pcs</i>	<i>4</i>	<i>200</i>
<i>Angle bar</i>	<i>2 pcs</i>	<i>2</i>	<i>600</i>
<i>Welding rod</i>	<i>1kilo</i>	<i>1</i>	<i>120</i>
<i>paint</i>	<i>1L</i>	<i>1</i>	<i>120</i>
<i>Bolt and nut</i>	<i>4pcs.</i>	<i>4</i>	<i>150</i>
<i>Plain sheet</i>	<i>1 pc</i>	<i>1</i>	<i>500</i>
<i>total</i>			<i>6,890</i>